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ORIGINAL ARTICLE

Evaluation of the antinociceptive effect and standardization of *Platycladus orientalis* (L.) Franco extract

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Abstract

Background *Platycladus orientalis* (L.) Franco (PO) has been used in Oriental medicine to treat rheumatism and diarrhea; however, its antinociceptive effects remain unclear.

Objective This study aimed to reveal the antinociceptive effect of the PO extract and standardize this extract in RAW 264.7 cells and mice with monosodium uric acid (MSU)-induced pain.

Results PO extracts were prepared using different ethanol (EtOH) concentrations (0–100%). Thereafter, groups were determined based on the results of the MSU-induced pain model in mice and nitric oxide (NO) assay using lipopolysaccharide (LPS)-induced RAW 264.7 cells. As the methylene chloride (CH_2Cl_2) and ethyl acetate (EtOAc) layers exhibited increasing effects with time, these layers were selected to isolate the major components. The PO extract was standardized using a highly efficient HPLC method, and its antinociceptive effects in a mouse model of MSU-induced gout pain were determined using the von Frey test. The PO extract was found to exhibit antinociceptive effects. Moreover, sample standardization was successfully achieved based on the validation of these compounds.

Conclusion The PO extract exhibited potent dose- and time-dependent antinociceptive effects in an animal model of gout pain; these effects might be mediated by the major compounds of this extract. Collectively, these results suggest that PO can be used as a herbal drug to treat gout-related pain.

Keywords *Platycladus orientalis* (L.) Franco · Phytochemical analysis · Pain · Antinociceptive effect · Standardization

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Introduction

Pain, one of the most prevalent conditions that limit productivity (Higgs et al. 2013), is associated with a physiological process called nociception, which is detected by a subpopulation of receptors called nociceptors (Regalado et al. 2017). Pain is associated with several disorders, such as inflammation, and requires analgesic treatment (Fongang et al. 2017). Chronic inflammatory diseases, such as rheumatoid arthritis, gouty arthritis, and autoimmune diseases, are currently the most predominant health problems worldwide, and chronic inflammation enhances pain sensitivity (Hwang et al. 2018). Among these conditions, gouty arthritis has become a common form of inflammatory arthritis owing to its increasing incidence and prevalence worldwide (Kodithuwakku et al. 2013).

Various candidates have been studied for the treatment of pain, including S-ketamine for neuropathic pain (Han et al. 2023) and sodium aescinate for relieving bone cancer pain (Yang et al. 2022). However, the options for acute and chronic pain treatment are limited, and many drugs are marginally effective with accompanying side effects (Ibrahim et al. 2017). In pain therapy, non-steroidal anti-inflammatory drugs and opioids are widely used, but side effects such as gastrointestinal damage, cardiovascular complications, tolerance, and physical/psychological dependence occur because of the prolonged use (Saldanha et al. 2017). Recently, interest in herbal medicinal products as a source in the search for new drugs is growing (Fongang et al. 2017), and compounds with significant analgesic properties in traditional medicinal plants are used to relieve pain associated with various diseases (Gorzalczany et al. 2011).

Platycladus orientalis (L.) Franco (PO) is oriental medicine plant recorded in the ancient Chinese material medica works in "ben Cao Bei Cao" (Fan et al. 2012). This plant has been used for gout, rheumatism, and diarrhea treatment (Lu et al. 2006). In addition, some Asian countries consider PO as a good source of healthy components and identified health benefits include antioxidant, neuroprotective, and anti-aging (Ren et al. 2017).

Plant standardization is a major challenge in the development of quality control measures to ensure effectiveness and safety. The benefit of standardization is related to the stable ingredients and yield of plants during extraction (Bezerra et al. 2017). Scientific methodologies are required for the validation and use of a product from traditionally used plant sources (Sahil et al. 2011). The flavonoid constituents of PO mainly include rutin, quercitrin, quercetin, amentoflavone, aroadendrin, myricetin, and hinokiflacone (Chen et al. 2007). However, the antinociceptive effects of PO and its major components remain unclear.

This study aimed to evaluate the antinociceptive effects of PO and standardize PO samples using isolated compounds in a mouse model of monosodium uric acid (MSU)-induced pain and the inflammatory responses induced by lipopolysaccharide (LPS) in RAW 264.7 cells.

Material and methods

Plant material

PO was harvested from Youngcheon-si in Gyeongsangbuk-do Province, South Korea. The specimen (NPLL-2144) was supplied by the Regional Innovation Center, Hallym University, South Korea.

Chemical reagents

All chemicals were of analytical grade. Methanol (MeOH) was acquired from J. T. Baker Chemical Company (Phillipsburg, NJ, USA), and trifluoroacetic acid and lipopolysaccharide (LPS) were purchased from Sigma-Aldrich Chemical Company (St. Louis, MO, USA). Dulbecco's modified Eagle's medium (DMEM), fetal bovine serum (FBS), and penicillin (100 U/mL)/streptomycin (100 µg/mL) (P/S) were purchased from Gibco (Grand Island, NY, USA).

Extract and standard solution preparation

PO (3 kg) was extracted at 27 °C using ethanol (EtOH) concentrations ranging from 0 to 100% for 4 days. After the solvent was evaporated *in vacuo* at 37 °C, the residue was dissolved in water and systemically partitioned into five fractions: hexane (*n*-Hex), methylene chloride (CH_2Cl_2), ethyl acetate (EtOAc), butanol (*n*-BuOH), and water (Fig. 1). All samples were analyzed in triplicate. Quercitrin and amentoflavone were dissolved in MeOH to prepare stock solutions, which were further diluted to prepare standards ranging from 9.8–490 µg/mL to construct calibration curves. Stock and working solutions were stored at 4 °C.

Phytochemical analysis using high-performance liquid chromatography (HPLC)

The chromatographic analyses were performed using an HPLC system (Dionex, CA, USA). The separations were conducted under gradient conditions using an Agilent Eclipse XDB-phenyl column (4.6 µm × 15 cm, with 3.5 µm particle size; Santa Clara, CA, USA) at 30 °C. The mobile phase consisted of water with 0.1% trifluoroacetic acid (TFA) (A) and MeOH (B). The following elution program was employed: 0–20% B (0–3 min), 20–30% B (3–12 min), 30% B (12–17 min), 30–87% B (17–23 min),

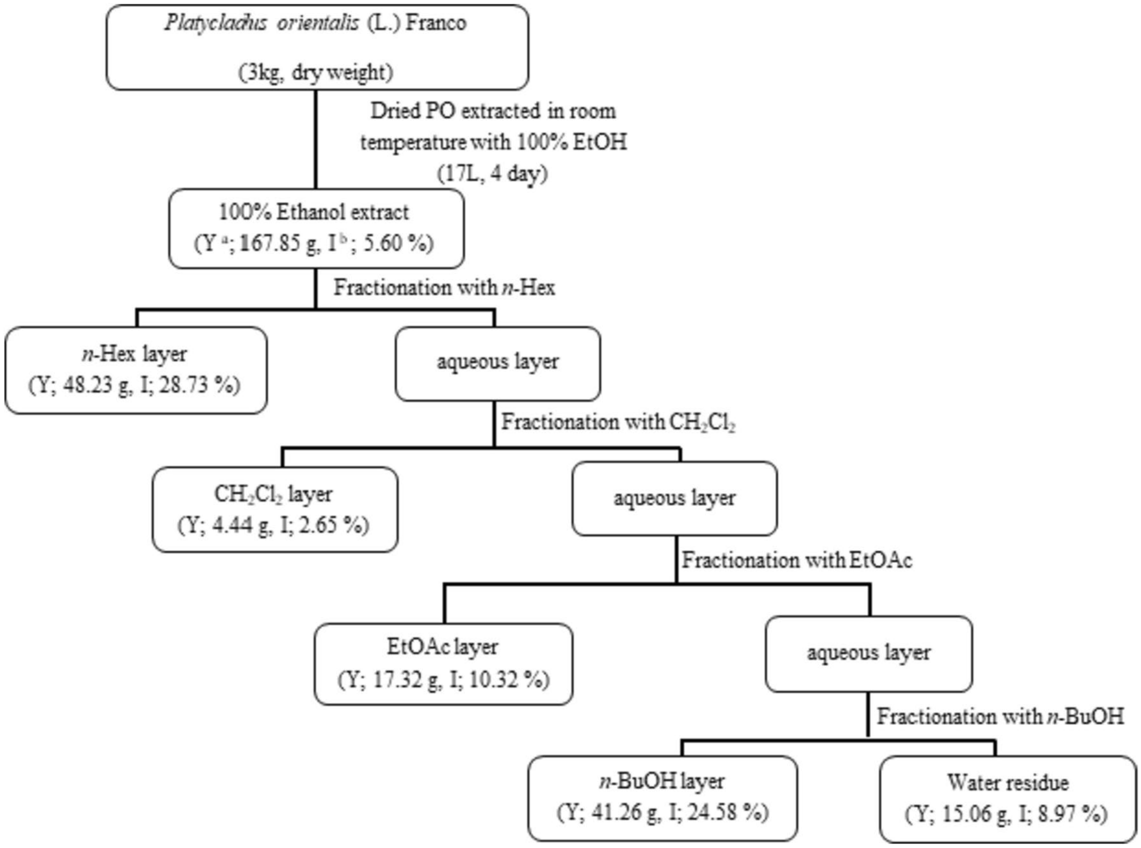

Fig. 1 Extraction and fractionation procedure for *Platycladus orientalis* (L.) Franco (PO) extract. [a]Yield, [b]inhibition (%, concentration: 50 µg/mL)

87–90% B (23–32 min), 90–100% B (32–35 min), 100–0% B (35–38 min), and 0% B (38–48 min). Peaks were detected using an ultraviolet (UV) detector at 254 and 280 nm. The flow rate was 0.7 mL/min, and the injection volume was 10 µL.

Animals

Animals for the von Frey filament test

Six-week-old male ICR mice (6 weeks old; Koatech, Seoul, Korea) were used for the experiments. Mice (5 per cage) were housed in a room at 22 ± 1 °C with alternating 12 h light–dark cycles. Food and water were provided ad libitum. The animals were allowed to adapt to the laboratory for at least 2 h before the initiation of testing and were only used once. To reduce variation, all experiments were performed during the light phase of the cycle (10:00–17:00). This study was approved by the Hallym University Institutional Animal Care and Use Committee (Registration Number: Hallym 2018–71) in accordance with the Guide for the Care and Use of Laboratory Animals published by the National Institutes of Health and the ethical guidelines of the International Association for the Study of Pain.

MSU-induced gout pain model

MSU crystals were prepared and injected as previously described (Hwang et al. 2018). MSU crystal was dissolved in phosphate-buffered saline (PBS) to a final concentration of 50 µg/µL and injected intra-articularly into one ankle joint. PBS alone was injected into the contralateral ankle joint under isoflurane anesthesia. Nociception was assessed 2 weeks after the injection of MSU.

von Frey filament tests

Five animals from each group were used for the von Frey test. The tactile withdrawal threshold of mice was measured using the von Frey method (Bonin et al. 2014; Chaplan et al. 1994). The mice were individually placed in wire-mesh floor cages (8 × 8 cm) to allow insertion of the mechanical probe from below against the plantar surface of the hind paw. The animals were allowed to acclimate to the testing chambers for at least 1 h before the first measurement. After the acclimation period, the plantar surface of the left hind paw was stimulated vertically using a series of von Frey filaments (North Coast Medical, Inc., Gilroy, CA, USA) with logarithmically increasing stiffness. The filament was bent to the central plantar surface with sufficient force for 5 s, and brisk withdrawal or paw flinching was considered a positive response. The tactile withdrawal threshold test was repeated twice for each mouse, and the mean value was calculated. Mechanical hyperalgesia was assessed at 0, 30, 60, and 120 min after oral administration of normal saline (5 ml/kg, b.w.), the PO ethanolic extract (50 mg/kg, b.w.), fractions (50 mg/kg, b.w.), and compounds (10 and 20 mg/kg b.w.).

Cell culture and treatment

Raw 264.7 murine macrophages were cultured and maintained at 37 ℃ under an atmosphere of 5% CO_2 in Dulbecco's modified Eagle's medium (DMEM) containing 10% heat-inactivated fetal bovine serum (FBS) with 1% (v/v) penicillin (100 U/mL)/streptomycin (100 µg/mL). Then, 1×10^6 cells/mL were cultured in a 24 well plate, and then pretreated with varying concentrations (0–50 µg/mL) of the extract for 1 h, followed by lipopolysaccharide (LPS) (5 ng/mL), and the supernatant was collected after 24 h.

Nitrite measurement

RAW 264.7 murine macrophages were treated with the extracts for 1 h, followed by LPS for 24 h. The culture media were collected and centrifuged at 12,000×*g* for 5 min, and the supernatant was collected to measure the nitrite level, which is an indicator of nitric oxide (NO) production. The supernatant (100 µL) was incubated with the same volume of Griess reagent (Sigma-Aldrich Chemical Company, CAT: G4410) for 20 min at room temperature in the dark. The absorbance of the reaction mixture was measured at 540 nm using a Spectramax M2/e fluorescence microplate reader (Molecular Devices, San Jose, CA, USA). Sodium nitrite ($NaNO_2$) was used as the standard to calculate the nitrite concentration.

Method validation

The formula in the International Conference on Harmonization (ICH) guideline Q2B (Sistla et al. 2005) was used to validate the linearity, limits of detection (LODs), limits of quantification (LOQs), intra- and inter-day precision, and recovery using HPLC. The LOD and LOQ were calculated at signal-to-noise ratios (S/N) of 3 and 10, respectively, using the standard deviation (SD) of the response and slopes of the concentrations obtained from the calibration curves. Intraday precision was evaluated using three replicates on the same day, whereas inter-day precision was assessed by comparing six replicates over three consecutive days. Three replicate extracts at each level were used to determine the extraction recovery rates, presented as means ± SD. Thereafter, the relative SD (RSD) was calculated.

Statistical analysis

All values are expressed as means ± SD. Comparisons between groups were performed using Student's *t*-test, and statistical significance was set at $p < 0.05$.

Results

HPLC analysis and effect on MSU induced pain model of PO extract

The HPLC chromatograms of PO extracted using different concentrations of EtOH are shown in Fig. 2. As depicted in Fig. 3, the 40% and 100% EtOH extracts exhibited higher antinociceptive effects than the other extracts for up to 120 min. Further, the effects of these extracts were significantly different at the end of the experiment.

Anti-inflammatory activity of PO extracted with different EtOH concentrations

As inflammation and NO generation are important factors that cause pain, the suppression of NO synthesis was assessed. The inhibitory effects of the PO extracts against LPS-induced NO release in RAW 264.7 cells differed according to the EtOH concentrations (Fig. 4). Furthermore, the 90% and 100% EtOH extracts exhibited the most potent suppression of NO synthesis. Therefore, the 100% EtOH extract was used in the von Frey test to evaluate the antinociceptive effects of PO.

HPLC analysis and effect of the 100% EtOH PO extract and fractions on the MSU-induced pain model

The HPLC chromatograms and antinociceptive effects of each fraction of the 100% EtOH PO extract are shown in Figs. 5 and 6, respectively. The CH_2Cl_2 and EtOAc fractions showed increasing time-dependent effects up to 120 min. In addition, the major peaks of the PO extract were detected in the CH_2Cl_2 and EtOAc fractions; therefore, the active compounds were isolated from each layer. Quercitrin was isolated from the EtOAc layer using a Sephadex-LH 20 column with 70% EtOH, and amentoflavone was isolated from the CH_2Cl_2 layer using silica gel.

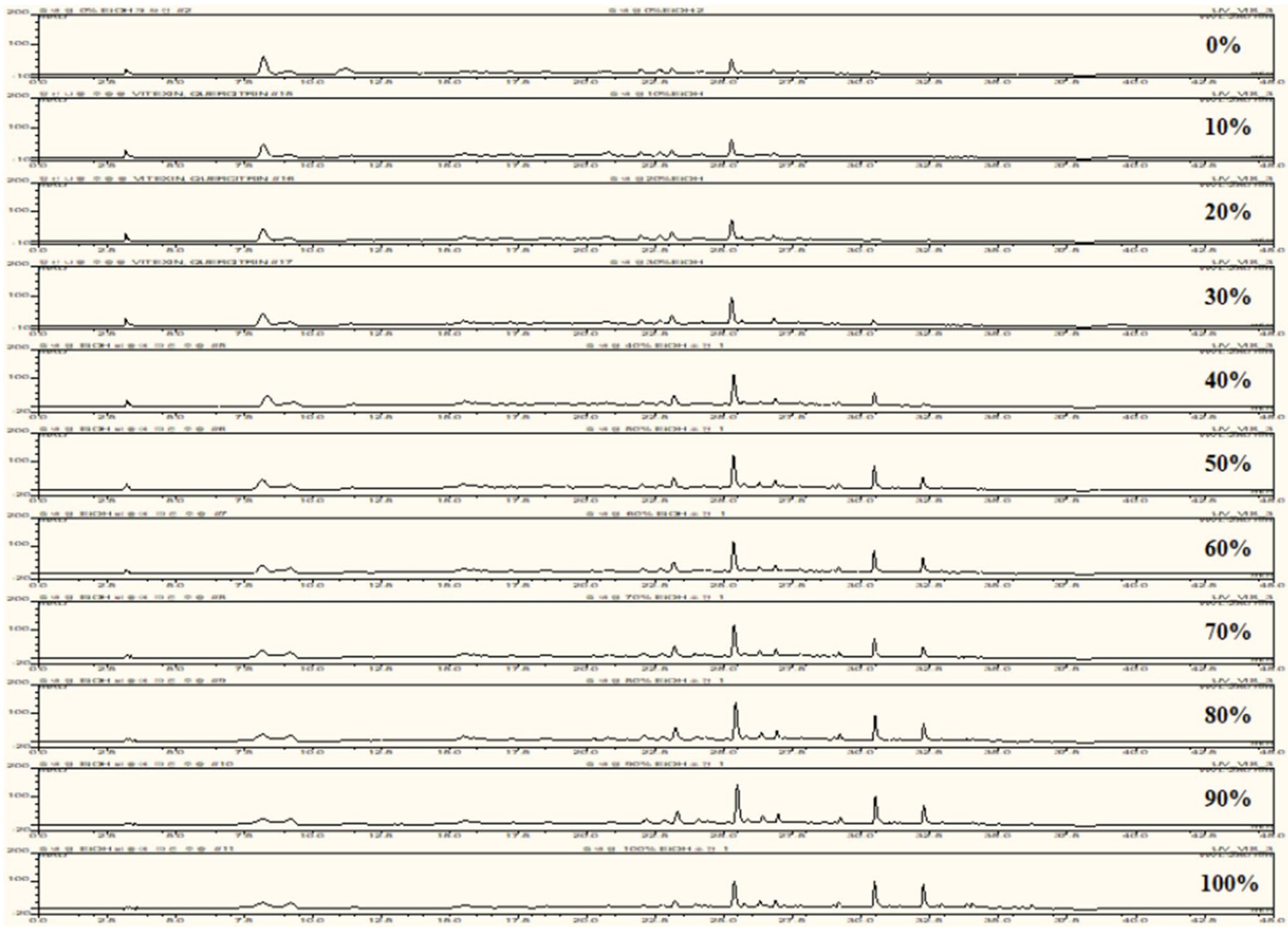

Fig. 2 High-performance liquid chromatography (HPLC) chromatograms (280 nm) of *Platycladus orientalis* (L.) Franco (PO) extracts prepared with different ethanol concentrations (0–100%)

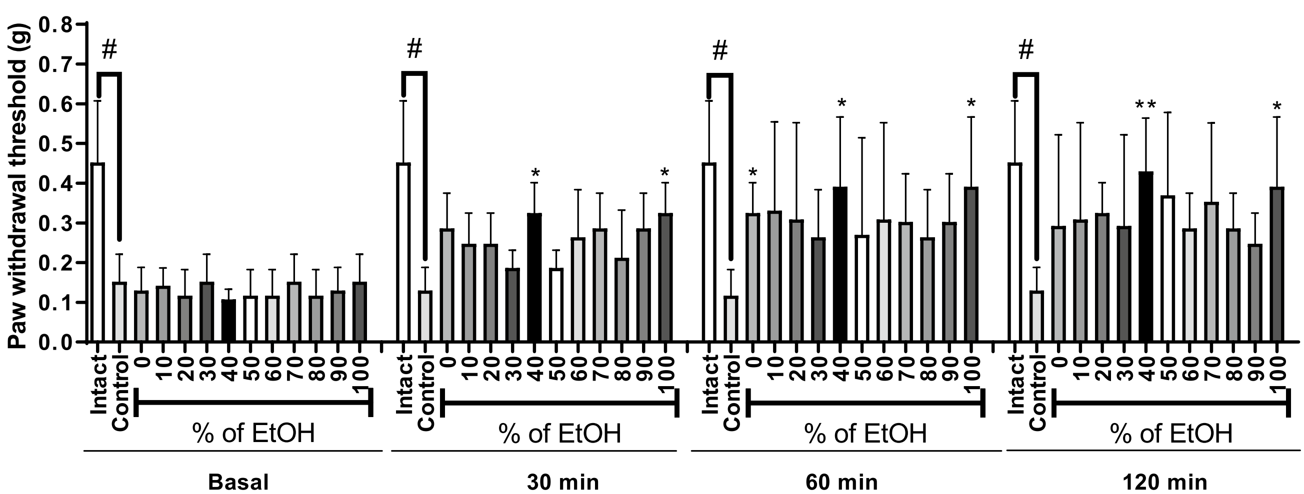

Fig. 3 Antinociceptive effect along different ethanol ratio extract from *Platycladus orientalis* (L.) Franco. Data are expressed as mean ± SD; n = 5 mice per group ($^{\#}p < 0.05$ *vs.* intact group; $^{*}p < 0.05$, $^{**}p < 0.01$ *vs.* control group)

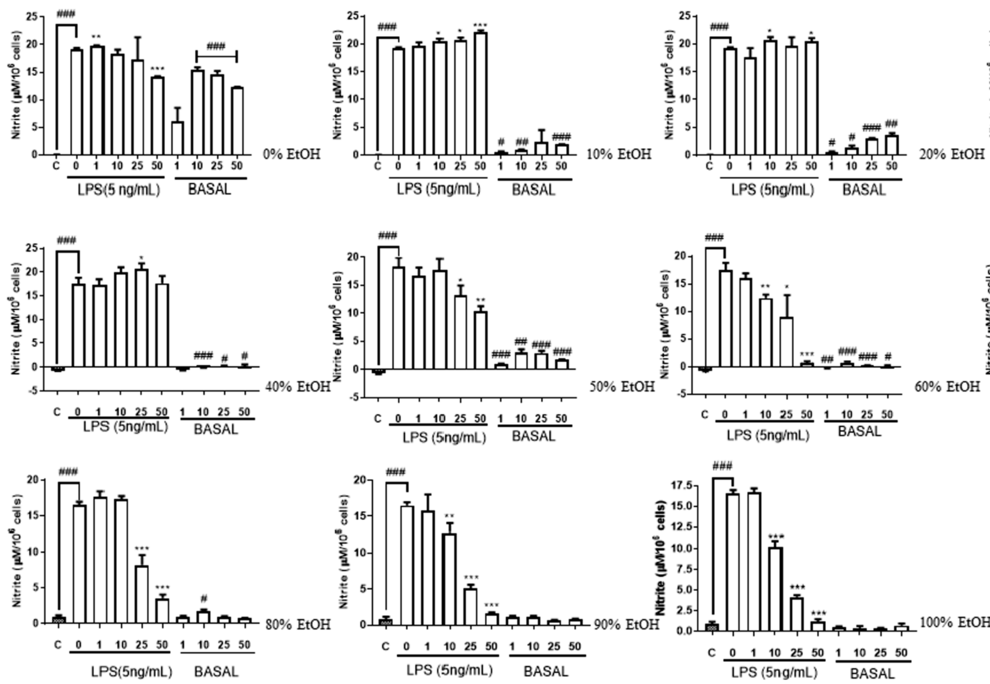

Fig. 4 Effect of extracts of *Platycladus orientalis* (L.) Franco along different ethanol ratio on LPS-induced NO release from cultured RAW 264.7 cells. Data are expressed as mean ± SD ($^{\#}p<0.05$, $^{\#\#}p<0.01$, $^{\#\#\#}p<0.001$ *vs.* the control group; $^{*}p<0.05$, $^{**}p<0.01$, $^{***}p<0.001$ *vs.* LPS treated group)

Validation

Optimization of the HPLC conditions

The major components of the PO extract were successfully separated using HPLC (Fig. 7). Various conditions, such as column, temperature, mobile phase, flow rate, and wavelength, were examined to develop an accurate, valid, and optimal HPLC method. Optimal separation was achieved using an Agilent Eclipse XDB-phenyl column (4.6 μm × 15 cm, with 3.5 μm particle size) with a mobile phase composed of 0.1% TFA in water and MeOH. Both selectivity and specificity were achieved using the established conditions.

Linearity (calibration curves), LOD, and LOQ

The linearity of the standard compounds was analyzed in triplicate at five concentrations. Calibration curves were prepared using the peak area versus concentration of the standard compounds. The developed method was linear across a concentration range of 9.8–490 μg/mL for both quercitrin and amentoflavone ($R^2 \geq 1.000$, Table 1). The LOQ and LOD were expressed as S/N ratios of 3 and 10, respectively, under the present chromatographic conditions. The LOD and LOQ for the standard compounds were 7.12 and 21.58 μg/mL (quercitrin), and 8.31 and 25.19 μg/mL (amentoflavone), respectively.

Precision, accuracy, and recovery

To validate the precision of the method, intra- and inter-day variations were analyzed. As shown in Table 2, the RSD (%) of the standard compounds were within the range of 0.11–1.22% (intra-day) and 0.20–1.33% (inter-day). Therefore, HPLC could be used to quantitatively determine the standard compounds in PO extracts with high precision. Recovery tests were performed using three concentrations of standard compounds, which were added to known samples. The accuracy of the method was proven based on the average recovery (%) results (Hwang et al. 2018), which ranged from 104.27–107.98% and 96.36–109.26% for quercitrin and amentoflavone, respectively (Table 3).

Effect of standard compounds from the PO extract on the MSU-induced pain model

The two concentrations (10 and 20 mg/kg) of quercitrin and amentoflavone led to a significant time-dependent increase in the antinociceptive effect for up to 120 min. Further, both compounds were highly effective when administered at higher doses (20 mg/kg) (Fig. 8). Of note, quercitrin

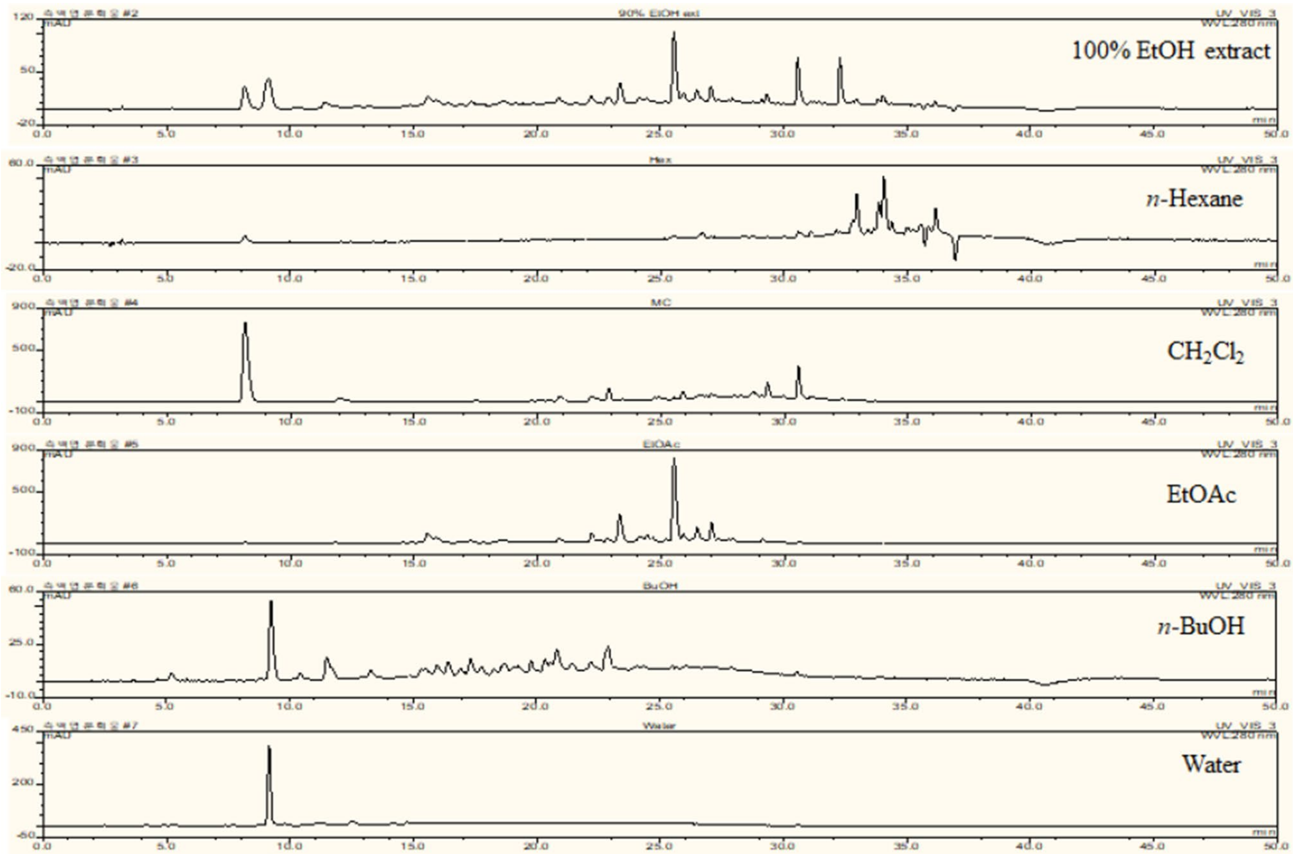

Fig. 5 Representative high-performance liquid chromatography (HPLC) chromatograms of fractions of 100% ethanol (EtOH) extract of *Platycladus orientalis* (L.) Franco (PO) at 280 nm

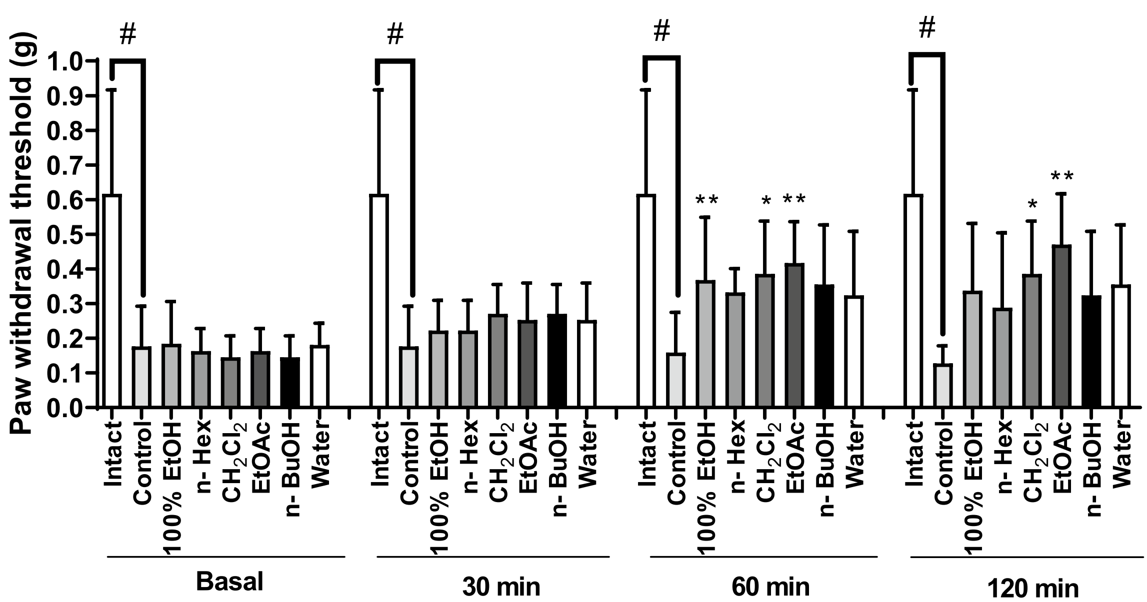

Fig. 6 Antinociceptive effect of 100% EtOH extract and fractions from *Platycladus orientalis* (L.) Franco. Data are expressed as mean ± SD; n = 5 mice per group ($^{\#}p < 0.05$ *vs.* the intact group; $^{*}p < 0.05$, $^{**}p < 0.01$ *vs.* control group)

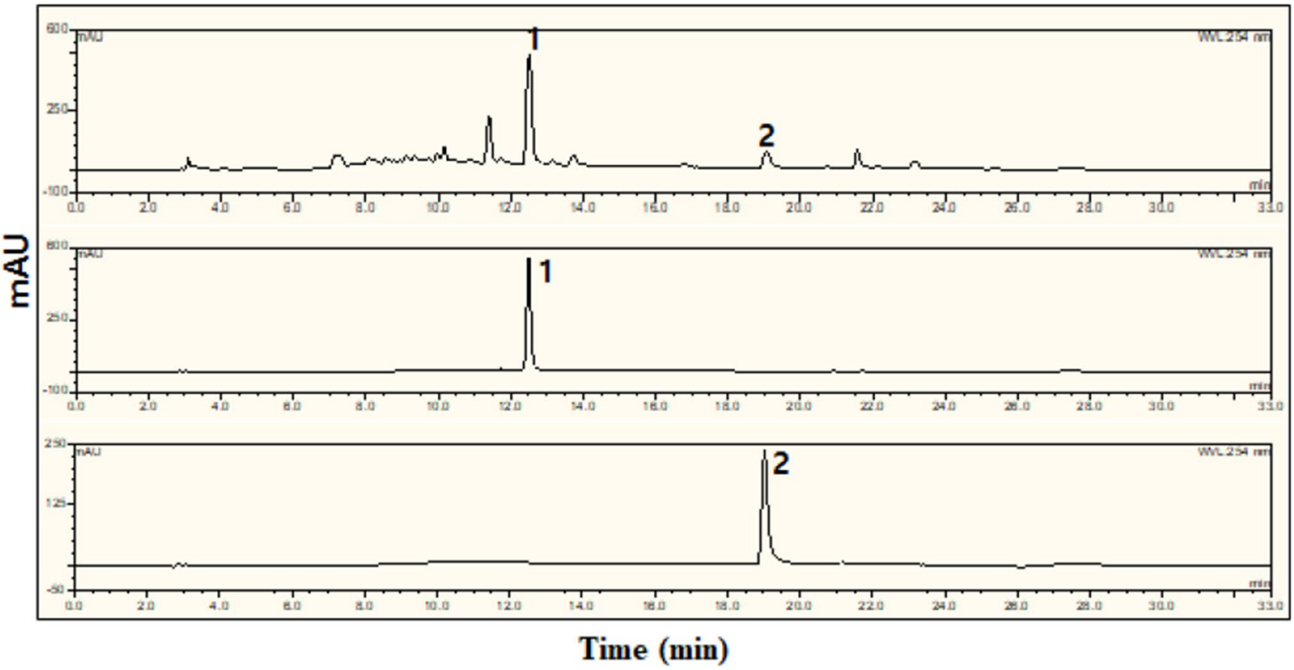

Fig. 7 Representative high-performance liquid chromatography (HPLC) chromatograms of 100% ethanol (EtOH) extract of *Platycladus orientalis* (L.) Franco (PO) and standard compounds at 254 nm; (1) quercitrin and (2) amentoflavone

Table 1 Method validation parameters for quantification of quercitrin and amentoflavone

Parameters	Quercitrin	Amentoflavone
Retention time (min; $n=5$)	12.05 ± 0.01	18.48 ± 0.02
Regression equation	$Y=0.5475X+1.0217$	$Y=0.5203X—2.3003$
Correlation coefficient (R^2)	1.000	1.000
Linear range (μg/ml)	9.8–490	9.8–490
LOD (μg/ml)	7.12	8.31
LOQ (μg/ml)	21.58	25.19

LOD: 3.3 × (standard deviation [SD] of response/slope of calibration curve), LOQ: 10 × (SD of response/slope of calibration curve), *LOD* limits of detection, *LOQ* limits of quantification

exhibited stronger antinociceptive effects than amentoflavone in an MSU-induced pain model.

Discussion

Based on the findings of this study, the PO ethanolic extract exhibited antinociceptive effects, which might be mediated by the presence of quercitrin and amentoflavone in a mouse model of MSU-induced pain. Mice were employed to establish a model of gout pain (Feng et al. 2019), and treatment with MSU was confirmed to reduce the pain threshold, as measured using the von Frey filament test. Gouty arthritis is characterized by MSU accumulation in the joints

Table 2 Intra-day and Inter-day precision of quercetin and amentoflavone

Compounds	Concentration (μg/mL)	Intra-day[a] ($n=3$)		Inter-day[b] ($n=9$)	
		Mean ± S.D	%RSD	Mean ± S.D	%RSD
Quercitrin	9.8	151.77 ± 0.17	0.11	152.13 ± 0.31	0.20
	49	192.74 ± 1.58	0.82	192.18 ± 2.32	1.21
	98	243.74 ± 0.76	0.31	244.64 ± 1.56	0.64
Amentoflavone	9.8	40.30 ± 0.49	1.22	39.20 ± 0.52	1.33
	49	76.70 ± 0.70	0.91	78.74 ± 0.21	0.27
	98	136.70 ± 0.25	0.18	137.45 ± 1.03	0.75

[a]Intra-day precision tested three times on 1 day

[b]inter-day precision tested on 3 consecutive days; RSD (%), relative standard deviation

Table 3 Recovery of quercitrin and amentoflavone

Compounds	spiked (μg/mL)	Theoretial (μg/mL)	Found (μg/mL)	Recovery (%)
Quercitrin	9.8	151.17	151.95 ± 0.25	107.98 ± 2.60
	49	190.37	192.46 ± 0.40	104.27 ± 0.81
	98	239.37	244.19 ± 0.63	104.92 ± 0.65
Amentoflavone	9.8	39.80	40.25 ± 0.07	104.54 ± 0.72
	49	79.00	77.72 ± 0.74	96.36 ± 1.50
	98	128.00	137.08 ± 0.53	109.26 ± 0.54

and tissues, resulting in hyperuricemia, which can induce chronic diseases such as edema, type 2 diabetes mellitus, and hypertension (Kiyani et al. 2019). Furthermore, MSU crystal accumulation can trigger excruciating pain in patients with gout (Fischer et al. 2018).

Based on the literature search, PO has been traditionally used to treat gout. The results of the study revealed that the EtOH extract reduced nociception in an MSU-induced pain model (Lu et al. 2006). Of the extracts prepared with different EtOH concentrations, the 40 and 100% EtOH extracts exhibited effective anti-inflammatory activity, which was attributed to the high concentration of quercitrin in the 40% EtOH extract and the high concentration of amentoflavone in the 100% EtOH extract, as shown in the HPLC chromatogram. Among the fractions of the 100% EtOH extract, the CH_2Cl_2 and EtOAc exhibited increasing time-dependent effects. Nociceptive transmission is modulated by nitro oxide (NO), which has dual nociceptive and analgesic effects related to pain transmission (Cury et al. 2011). Based on this theory, the effect of the 40% EtOH extract on NO release suggests that NO has a different effect on inflammation. Further, the 40% EtOH extract exhibits other metabolism-related antinociceptive effects.

Studies on the active compounds in PO sought to identify components, including flavonoids (Chen et al. 2007), and purify active antioxidant and anti-inflammatory polyphenols (Ren et al. 2019). To determine the constituents of the PO extract that mediate its antinociceptive effects, the effective compounds, quercitrin, and amentoflavone, were isolated from the CH_2Cl_2 and EtOAc fractions. An HPLC method was developed to quantify the two standard compounds in the PO extract, and the results were successfully validated. The HPLC method established in this study exhibited adequate efficiency, which enabled its application in the quality assurance/quality control (QA/QC) analysis of the PO extract. To the best of our knowledge, this is the first study to standardize PO extracts based on quercitrin and amentoflavones using HPLC. In addition, the two standard compounds exhibited time-dependent antinociceptive effects at 10 and 20 mg/kg in the MSU-induced pain model; 20 mg/kg quercitrin exhibited stronger effects than the other compounds and doses. In previous studies, the antinociceptive effect of quercitrin was related to its free radical-scavenging activity (Hasan et al. 2006) and modulation of pro-inflammatory mediators (Gadotti et al. 2005). Anti-inflammatory (Kim et al. 1998) and antinociceptive effects have been examined in the management of neurodegenerative disorders (Ishola et al. 2013). These previous results corroborate those obtained in the present study, indicating that the two

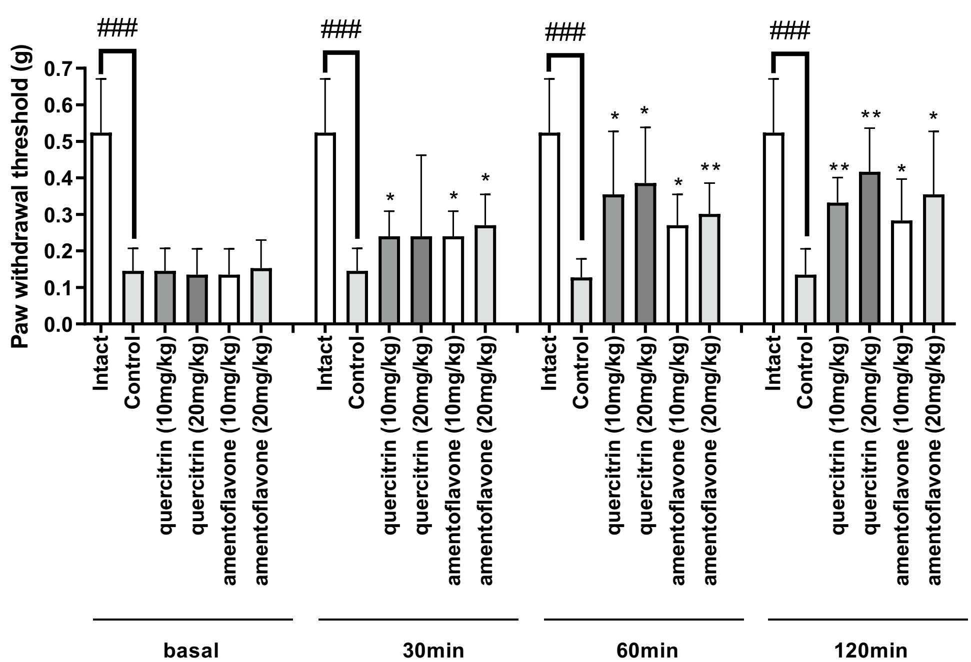

Fig. 8 Antinociceptive effect of compounds from *Platycladus orientalis* (L.) Franco. Data are expressed as mean ± SD; n = 5 mice per group ($^{###}$ $p < 0.001$ *vs.* the intact group; * $p < 0.05$, ** $p < 0.01$ *vs.* control group)

identified standard compounds affected the gout model and could be useful for the management of gout-related pain. Further studies are warranted to explore the exact mechanisms underlying the antinociceptive effects of PO.

Conclusions

The PO ethanolic extract exhibited antinociceptive effects in the MSU-induced gout pain model. The effect of the PO extract may have been mediated by its two main active compounds, quercitrin and amentoflavone, which exhibited dose- and time-dependent antinociceptive effects in the same animal model. The HPLC validation results suggested adequate efficiency of the two compounds. Overall, our findings may be valuable for the development of further strategies to acquire information on the antinociceptive activity of the PO extract and its major compounds that can be used as herbal drugs for the treatment of gout-related pain.

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Author contributions HYK, GZ and HJL performed experimental work. SHH prepared traditional plant sample for experiment. HYK, GZ and SKL analyzed the data. HWS and SSL designed the study and supervised them. HYK prepared the manuscript for publication, and JHP and SSL reviewed the draft of the manuscript. All authors read and approved the final manuscript.

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Data availability statement Not applicable.

Declarations

Conflict of interest Author Hyun-Yong Kim declares that he has no conflict of interest. Author Guanglei Zuo declares that he has no conflict of interest. Author Hee Jung Lee declares that she has no conflict of interests. Author Seung Hwan Hwang declares that he has no conflict of interest. Author Soo Kyeong Lee declares that she has no conflict of interest. Author Jun Hong Park declares that he has no conflict of interest. Author Hong-Won Suh declares that he has no conflict of interest. Author Soon Sung Lim declares that he has no conflict of interest.

Ethical approval Animal protocols were approved by the Hallym University Institutional Animal Care and Use Committees (Registration Number: Hallym 2018–71).

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